

## 1. Problem Statement and Target Users

- What real-world problem does your application aim to solve?

The aim is to solve problems related to the healthcare sector, The healthcare sector is known to be busy and operates under enormous strain as it is driven by high patient demand, particularly from non-emergency patients. Due to this, hospitals must absorb double the patients needed to care, leading to long wait times. The application is to help increase efficiency on administrative tasks so that hospitals can clear non-emergency patients faster, while it is able to spend more time on emergency patients with critical conditions.

There are some underlying symptoms that may or may not happen immediately after the patient had it, and it might be minor or major symptoms. The application is useful to help those that need medical care and it helps to reduce the demand for non-emergency situations.

- What are the intended users of the application?
  - Those who are hesitant to or are unsure about their symptoms warrant the need to see a doctor. The application would provide an initial assessment and help determine the next step for them.
  - Nurses to determine where they stand in the queue among other patients; they have so that the ones that urgently need it first can be prioritized.

## 2. User Inputs

- What information or data will users provide to the system?
  - Patient or Nurses
  - Symptoms
  - Symptoms history and time stamp
  - Name
  - NRIC (Nurses)
  - Age
  - Gender
  - Drug Allergy

## 3. Use of AI

- How will AI be utilized within the application?
  - AI will be used to process information given by the user and determine what to suggest to the user on what to do next.

- It will ask follow-up questions and such to gain more information before deciding
- It will give the severity of symptoms whether it requires medical care
- What outputs, insights, or recommendations will the AI generate from the user inputs?
  - The AI will take the symptoms and form an initial diagnosis and recommend the appropriate next step for them, whether it is serious enough to go see a doctor or to wait out and monitor.
  - The AI will keep a log which the user can print or email the log to send to the healthcare provider to aid them better.

#### 4. Business Rules

- What business rules, validations, or decision-making logic will be applied to the AI-generated outputs?
  - AI Failure handling
    - If the AI service fails, it will display an error message and ask the user to reload page or wait for a few minutes if the error persists.
  - Emergency symptom detection
    - difficulty breathing and severe bleeding. It will prioritize recommending getting emergency medical aid.
  - Severity weightage
    - Different levels of urgency like urgent or un-urgent
  - Contradictory or incomplete information.
    - Will prompt user for more information if information given is unclear or incomplete
  - Uncertainty and confidence handling
    - Will ask follow-up questions instead of making assumptions
    - Will recommend a consultation if it cannot determine condition
  - Able to filter important information can be displayed for user to take note of
    - History of how the user got the injury or symptom
    - History of what solutions the AI can give
  - Recommendations given to user
    - If it is an injury eg a cut, the application will give steps how to remediate it instead of letting the user figuring it out by himself